

Ryan Hsu

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Education

University of California, Berkeley – B.A. Data Science and B.A. Public Health 2022 - Present

Coursework: Computer Science (Python/Java focus), Data Science(AI/ML focus), Statistics, Public Health, Biology

Certifications: AWS Certified AI Practitioner

Projects

Pacman AI Game

- Implemented classic and informed search in Python (DFS, BFS, Uniform Cost, A* with admissible/consistent heuristics) to solve maze navigation, improving pathfinding efficiency and solution reliability across large maps.
- Developed multi-agent controllers using Minimax and Expectimax with alpha-beta pruning, designing domain-specific evaluation functions to handle stochastic ghost behavior, and increasing win rate 8x (10% to 80%) against baselines.
- Trained reinforcement-learning agents (Q-learning and approximate Q-learning) and built probabilistic inference pipelines (Bayes Nets, HMM particle filtering) to localize hidden ghosts from noisy sensors, packaging the work with a CLI, unit tests, and lightweight CI for repeatable runs.

Hume AI Poker Bot

calhacks-hume-poker.vercel.app

- Created a bot that utilizes Hume AI's Expression Management API to detect key emotions that can dictate poker performance and skill level.
- Utilized ChatGPT API to advise new players on ideal ways to play poker hands using probability and game state.

Java-Based 2D Game

- Designed and implemented a 2D tile-based world exploration engine in Java, creating randomly generated and visually distinct worlds, enhancing user experience by ensuring unique layouts and seamless navigation.
- Developed and integrated user interactivity features, including avatar movement, HUD elements, and save/load functionality, resulting in a consistent and engaging user experience.

Professional Experience

Software Engineer Intern, HopeKCC

June 2025 – Sept 2025

- Collaborated on an open-source, cross-platform music synchronization app enabling real-time display of ChordPro scores, developed by interns across web, mobile, backend, and QA teams under Agile workflows.
- Contributed to the development of a responsive web admin dashboard for managing ChordPro-formatted music libraries, implementing frontend features using modern frameworks, and integrating with backend APIs.
- Assisted with Google Cloud Platform (GCP) deployment of production-grade Flask services to Cloud Run, resolving containerization, IAM, and service routing issues in a CI/CD pipeline.

Web Development Associate, Ethical Apparel

Sept 2023 – Present

- Revamped the custom orders page, creating a more streamlined and user-friendly system for placing custom orders.
- Managed the online storefront for Ethical Apparel, attracting 10,000+ unique visitors annually.
- Developed and implemented new sales promotions, increasing customer engagement and sales.

Research Assistant, Yale School of Medicine

Summer 2022

- Recruited 20+ physicians for an international study on capnography trace scoring during facemask ventilation.
- Expanded field knowledge by collaborating closely with Dr. William Rosenblatt on the study.

Leadership Experience

Internal Vice President, Taiwanese American Student Association

Sept 2022 – Present

- Led programming for 100+ members; ran biweekly events and logistics across a 40+ person board.
- Managed budgets and payments on CalLink/Venmo; grew club finances by 125% and standardized reimbursements.

Skills

Programming Languages: Python, Java, JavaScript (HTML/CSS), SQL (PostgreSQL), R, Scheme

Libraries & Tools: TensorFlow, PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, XGBoost, Flask, Jinja2, REST/JSON, Git, Docker, GitHub Actions, GCP, Vercel, AWS, Hume Expression API, OpenAI/ChatGPT API

Languages: English, Mandarin Chinese (healthcare interpreting), Spanish